Omnipair Dusk: A Generalized Automated Market Maker for Oracle-less Lending and Leveraged Liquidity

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Expanded technical draft for Omnipair Dusk v2

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Abstract

Omnipair Dusk combines exchange, isolated lending, two-sided liquidity shares, and leveraged liquidity vaults in one market on Solana. Its accounting identity separates cash, indexed cash-backed debt, and vault-owned reserve claims. Swap execution applies a second separation: unpaid cash-backed interest is excluded from curve principal, and hLP ownership is removed to obtain the ordinary liquidity tranche on which traders execute. A full-range constant-product tail and two nested liquidity bands define a continuous, piecewise analytic reserve path. After a trade, the program reconstructs hLP ownership and opposite-asset funding algebraically. Credit uses conservative internal price and depth observations, while realized yield, recenter funding, and loss recovery have distinct accounting paths. This paper describes the implemented mechanisms and their limitations, with reproducible numerical illustrations of the curve, interest controller, and hLP recovery incentives.

Implementation scope. This revision follows Dusk source at commit `263c1635ad87` (September 10, 2026). Dusk is under active development and has not been deployed to production. Equations below suppress token normalization and integer rounding unless stated otherwise. The source map and generated examples accompany both drafts; numerical illustrations are not an audit, market backtest, or return forecast.

1 Introduction

An AMM, a lending market, and a liquidation route often hold separate views of the same asset. A lender may value collateral at a linear price even though recovery must occur through a finite pool. For a thinly traded pair, the difference between that mark and executable recovery can dominate the economics of the loan. External price feeds can improve information, but do not create the inventory required to liquidate a position.

Dusk develops Omnipair's Generalized Automated Market Maker (GAMM) approach: exchange and credit share a pair-local balance sheet [1]. Borrowing replaces available cash with a debt claim, while trading changes the inventory coordinate. The market also maintains two aggregate hLP vaults, each targeting exposure to one asset through a funded position in the

market's own liquidity shares. The intended benefit is a common accounting and settlement system for activities that otherwise depend on separate venues.

The current design has three important separations. First, stored reserve claims differ from cash available for a transfer. Second, the coordinate used to mint and burn liquidity shares differs from the ordinary tranche used to quote a swap. Third, collateral admission, liquidation eligibility, and actual liquidation proceeds use different valuations. These distinctions are necessary once an AMM contains debt, concentrated liquidity, and leveraged ownership at the same time.

"Oracle-less" means that the program does not require an external price feed to execute these mechanisms. It does use internal price observations, moving averages, depth measurements, and a persistent concentration center. Their quality depends on the market's trading history. Market-local accounting contains a loss within its accounting domain; it does not establish an external fair price or guarantee that liquidity remains available.

1.1 Related mechanisms

StableSwap demonstrated how an invariant can allocate more liquidity near an expected exchange rate [2]. CryptoSwap developed internal price adaptation and an economic budget for moving concentrated liquidity [3]. Dusk uses a different curve construction: a nonzero constant-product tail plus nested core and shoulder ranges, with an explicit custody bucket for retained recenter funding. The mathematical form and financing of that movement must therefore be evaluated on their own terms.

The adaptive rate-at-target idea is closely related to Morpho's AdaptiveCurveIRM [4]. Dusk implements its own bounded, discrete anchor update and accrues a shared asset-side borrow index. Similar controller intuition does not imply identical integration, parameters, or numerical behavior.

Leveraged liquidity has a separate lineage. Egorov's treatment of constant-product LP convexity motivates one-sided exposure through dynamically maintained leverage [5]. Yield Basis holds Curve LP collateral against crvUSD debt; its LEVAMM and VirtualPool support the rebalance mechanism [6]. Dusk represents the LP ownership and funding inside the same market and reconstructs them at the quoted endpoint. Section 4 makes the scope of that distinction precise.

1.2 What is established here

The reserve identities and endpoint formulas are accounting statements. Their continuous forms can be derived and checked independently. Whether the Rust program preserves them under every token decimal configuration, overflow boundary, account combination, and instruction sequence is an implementation question. Whether a configured market offers useful credit and sustainable LP returns is an empirical question. The examples at the end address selected arithmetic properties, while the expanded appendices connect those properties to source and describe the remaining evaluation work.

2 Market Accounting

Consider base and quote assets, denoted b and q. For each side i, write C_i for accounted reserve cash, D_i^f for indexed public-borrow debt, D_i^ℓ for indexed isolated-leverage debt, and H_i for the aggregate hLP live-reserve component. The stored live reserve is

$$\boxed{R_i = C_i + D_i^f + D_i^\ell + H_i.} \tag{1}$$

Here $D_i^{\mathrm{cb}} = D_i^f + D_i^\ell$ is all cash-backed debt. Isolated leverage is a separate health domain, but is already included in cash-backed debt. hLP funding debt D_i^h affects utilization and vault equity; it is not added again to the cash-backed term.

Let P_i^f and P_i^ℓ be the outstanding principal counters. Unpaid cash-backed interest is

$$J_i = \left[D_i^f - P_i^f\right]_+ + \left[D_i^\ell - P_i^\ell\right]_+ . \tag{2}$$

The executable reserve coordinate before separating hLP ownership is

$$X_i = R_i - J_i. \tag{3}$$

Thus interest accrual can increase the stored live reserve without expanding the curve principal. Using R_q/R_b as a universal swap price would miss both this adjustment and concentrated-curve geometry.

2.1 Cash, custody, and liquidity shares

Accounted cash is only one claim on the physical reserve vault. Claimable swap fees, hLP backing inventory, and protected recenter reserves also occupy custody. The program requires physical tokens to cover their sum:

$$V_i^{\mathrm{reserve}} \;\geq\; C_i + F_i^{\mathrm{custody}} + B_i^b + B_i^q + Z_i, \tag{4}$$

where B_i^b, B_i^q are backing inventory assigned to the two hLP vaults and Z_i is protected recenter funding. An unsolicited token donation does not automatically increase executable liquidity. Interest-vault accounting is separate from this reserve-vault check.

yLP is a fungible, market-wide two-sided liquidity claim. It does not give each holder an individual tick range. For total yLP supply S and a burn of s shares, the proportional reserve amounts use the stored live basis:

$$a_i = \left\lfloor \frac{sR_i}{S} \right\rfloor . \tag{5}$$

The withdrawal must still pass cash, market-health, and slippage checks. The existence of a share claim is not an unconditional redemption promise. The program checkpoints published yield before changing eligible balances; a principal withdrawal does not replace a separate harvest of that yield.

2.2 Borrow and repayment

At the principal step of a cash-backed borrow of a, C_i decreases by a and debt increases by the corresponding indexed share amount. Ignoring rounding and simultaneous accrual, $\Delta C_i = -a$, $\Delta D_i^{\mathrm{cb}} = a$, and $\Delta R_i = 0$. This is borrow neutrality: replacing inventory with a loan claim need not itself trade through the curve. Integer aggregate share conversion can produce a small reserve adjustment, and the full instruction also refreshes market state.

For repayment r against debt D and principal P, the principal portion is $p = \lfloor rP/D \rfloor$, with exact principal clearance on full repayment. The remainder $\iota = r - p$ is realized interest. Principal returns to reserve cash; interest follows the interest distribution path. In the idealized accounting step, $\Delta R_i = -\iota$ and $\Delta J_i = -\iota$, so X_i is unchanged. Accruing a receivable and realizing yield are different events.

2.3 Why several reserve coordinates are necessary

Equation (1) records ownership claims, not a guarantee that all claims can leave custody at once. A public loan can remain outstanding while a trader obtains a quote, but the trader's output still needs a cash source. Similarly, an hLP live-reserve entry has an owner and a funding obligation, yet does not create an additional token in custody. A safety check on the ledger and a safety check on the transfer therefore answer different questions.

Equation (3) removes a further ambiguity. Unpaid interest belongs in the debt balance and influences risk, rates, and share accounting. Allowing it to create executable swap principal would quote a receivable as if it were newly supplied inventory. The program explicitly subtracts public and isolated unpaid interest before constructing its executable reserve state. The hLP transition separately accounts for hLP funding interest.

The total yLP supply includes the balances owned by both aggregate hLP vaults. During an integrated swap, the ordinary holders' share count remains fixed while the vault-owned balances may be reconstructed. It is therefore useful to distinguish S, total supply used in proportional ownership, from S_o, the ordinary share supply held outside those vaults. Treating them as interchangeable would obscure who receives a fee or absorbs a change in vault equity.

3 Concentrated Exchange and Yield

Trading occurs on an ordinary yLP tranche (U, V) obtained after removing hLP ownership from (X_b, X_q). Let $z = \sqrt{p}$ denote the curve's marginal quote-per-base price in normalized units. The continuous reserve path is a full-range tail of liquidity $L_0 > 0$ plus up to two nested ranges. For range j with liquidity L_j and square-root boundaries $a_j < b_j$, define

$\bar{z}_j = \text{clip}(z, a_j, b_j)$. Then

$$U(z) = \frac{L_0}{z} + \sum_j L_j \left(\frac{1}{\bar{z}_j} - \frac{1}{b_j} \right), \tag{6}$$

$$V(z) = L_0 z + \sum_j L_j (\bar{z}_j - a_j). \tag{7}$$

The inner range is the core; the wider range supplies the shoulder. In the core both are active. In either shoulder only the wider range is active. Outside both ranges, the full-range tail remains. The normal nested path has five regions separated by four boundaries.

Within one region, active liquidity L_A is constant. A base input ΔU moves the square-root price down according to

$$z' = \left(\frac{1}{z} + \frac{\Delta U}{L_A} \right)^{-1}, \qquad \Delta V = L_A(z - z'). \tag{8}$$

If an input reaches a boundary, the quote continues in the adjacent region. Quote input is symmetric: $z' = z + \Delta V / L_A$ and base output is $L_A(1/z - 1/z')$. This gives a bounded segment walk rather than an iterative invariant root solve. Geometry preparation still uses bounded integer arithmetic; the statement concerns the execution path across prepared segments.

The three shape controls are peak amplification, core half-width, and fade width. With a shoulder enabled, concentrated liquidity is divided between the inner and outer ranges, with integer rounding favoring the core. At amplification one and zero widths the curve is exactly constant product. In that special case $p = V/U$ and output is $\Delta U \, V/(U + \Delta U)$. In a concentrated region p is the marginal price obtained from geometry, generally different from the inventory ratio V/U.

3.1 Center movement and fees

The concentration center is persistent state. An eligible operation can commit a previously deferred controller target before preparing its quote. The new trade observation may schedule a later target; it does not first move the center using its own endpoint and then reprice itself against that center. Thresholds, an adjustment step, an interval, and economic admission checks constrain center movement.

The swap fee combines a base charge, an outward-divergence surcharge, and a volatility surcharge. The active path evaluates divergence over a gross trade path and limits the charge with component and aggregate budgets. It does not run the historical implicit fee solver. Volatility pricing uses decayed prior state, and a successful trade updates the signal used by later operations.

Fees can be collected on input, on base only, or on quote only. If the fee asset is input, the budget is measured on actual reserve credit after the input token transfer fee. If the fee asset is output, it is measured on gross curve output. In either case total Dusk swap fees are bounded by half of that fee basis, rounded down. Token-program transfer fees are additional; the bound is not a universal limit on the user's total loss from fees and price impact.

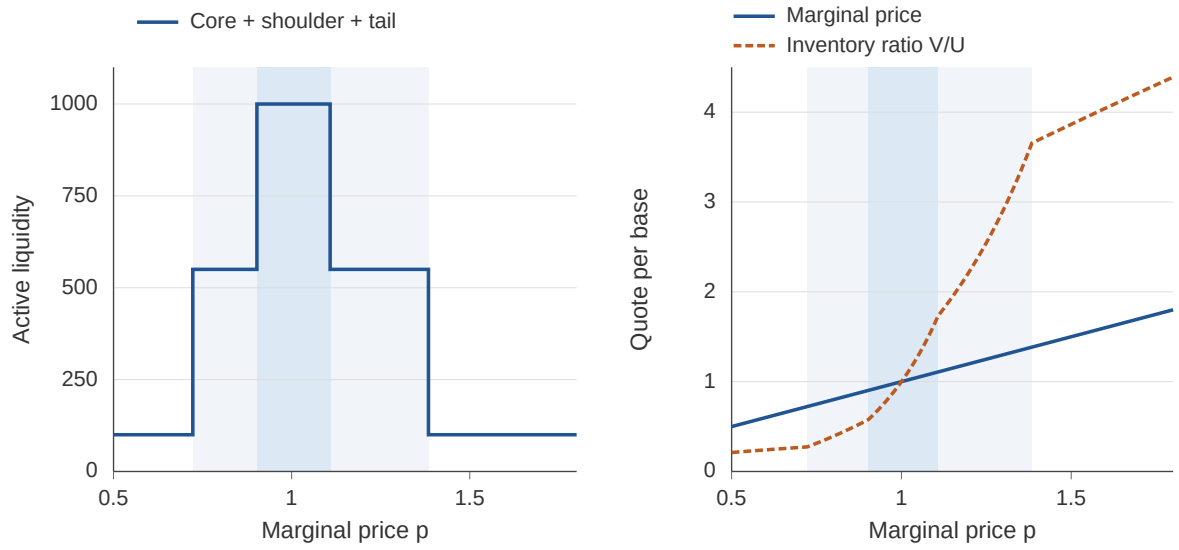


Figure 1: An illustrative fixed nested geometry. Marginal price and inventory ratio remain continuous across the boundaries, while active liquidity changes by region. The dark core and light shoulder describe the same two ranges in both panels. The example is specified in the companion data.

3.2 Claimable yield and compounding

A configured portion of eligible LP base fees and distributed dynamic surcharges can compound into principal. The remaining claimable swap yield is held outside executable reserves and distributed through growth indexes. Realized borrow interest follows its own distribution path. Thus the earlier description of all yield as universally non-compounding no longer applies.

For an eligible balance s, an index increment ΔG_i produces a yield increment of the form $s\Delta G_i$, with fixed-point scaling and fractional remainder handling in the implementation. Eligibility must be measured at the correct transition boundary: changing hLP ownership before allocating a fee could assign the trade's yield to a different set of shares.

Retained dynamic surcharge has another purpose. It enters Z_i in equation (4) and remains unavailable for swaps or harvest until an admitted recenter deploys it. It is not interchangeable with claimable yield, base-fee compounding, or unpaid interest. The controller's ability to move liquidity is therefore constrained by a specific funding source.

3.3 Launch controls and recipient authority

The fee profile includes an optional launch schedule. Its time decay can be linear or use a sixteen-step exponential schedule; an optional price-progress schedule also has a bounded duration. A directional size premium prices an individual qualifying buy. It is stateless across

transactions, so splitting an order across transactions can reduce the premium. While the directional limiter is active, the program rejects multiple same-market swap-like top-level actions in one transaction. These rules do not imply a universal per-wallet trading quota or resistance to all forms of order splitting.

Yield ownership, destination, and execution authority are distinct. The yield-account owner can set the recipient and an optional harvest authority independently. The owner, recipient, or configured harvest authority may invoke harvest, but payout goes to the recipient's canonical associated token account for the mint and token program. The additional authority authorizes execution of harvest; it does not give the signer arbitrary payout routing. An owner that is a program-derived address can act through its program's signed invocation.

Compounding changes a reserve claim, whereas harvest settles a published liability. The integrated fee transition allocates value to the pre-swap eligible yLP ownership, converts each hLP's allocation into its target equity, and reconstructs the matching funding position. This is why simply adding the entire fee to one raw reserve after a trade is not a complete description of the current fee path.

4 Leveraged Liquidity

The base hLP vault holds a funded yLP position whose net exposure targets base; quote hLP is symmetric. Let E_b, E_q be their target-asset equities after accounting for the funding and yield effects included in the transition. In the continuous model, a canonical endpoint is

$$X_b = U + E_b + E_q \frac{U}{V}, \qquad D_b^{h,*} = E_q \frac{U}{V}, \tag{9}$$

$$X_q = V + E_q + E_b \frac{V}{U}, \qquad D_q^{h,*} = E_b \frac{V}{U}. \tag{10}$$

The opposite-asset yLP claim and the target funding obligation match. For ordinary share supply S_o, vault-owned shares are reconstructed as

$$s_b = \left\lfloor \frac{S_o E_b}{U} \right\rfloor, \qquad s_q = \left\lfloor \frac{S_o E_q}{V} \right\rfloor, \qquad S = S_o + s_b + s_q. \tag{11}$$

These formulas describe aggregate vault state and require no scan over hLP holders.

The trader first receives a quote on (U, V). The program then reconstructs the two vaults at the quoted endpoint and verifies the associated debt, cash, interest, and reserve changes. Synthetic hLP inventory is not an extra free tranche of trader depth. The earlier fixed-point pre-rebalance, bisection, and tracking-loss-threshold description is not the active integrated mechanism.

For base hLP, its gross LP value at marginal price p is $E_b p + E_b V/U$, and debt value is $E_b V/U$. Relative to target equity value $E_b p$, gross leverage is

$$\mathcal{L}_b = 1 + \frac{V}{Up}, \qquad \mathcal{L}_q = 1 + \frac{Up}{V}. \tag{12}$$

Both equal two at a balanced constant-product point. With concentrated liquidity away from the center, inventory weights differ and a universal fixed-2× claim would be incorrect.

4.1 Funding and settlement constraints

An hLP deposit is admitted only if projected indexed funding debt fits the current borrowed-asset cash headroom. This is a deposit check, not a global theorem that funding debt remains below cash after every swap or after passive interest accrual. A spot swap first quotes full accrued hLP funding-interest settlement. If that payment would consume cash needed for trader output or the final reserve atom, it requotes without a funding-recovery bonus and carries both active vaults' unpaid funding interest on indexed debt. The hLP debt principal and synthetic borrowed reserve remain separate, so carrying interest cannot erase the liability or create spendable AMM principal. A later swap can pay the carried interest when cash is sufficient. Trader output must still be covered by actual reserve cash; other transition types retain their own cash requirements.

The implementation binds a prepared hLP transition to its expected pre-state. After applying the cash and funding effects, it checks the reconstructed executable endpoint within a fixed tolerance of three raw token atoms per side and asserts the stored reserve identity. Token normalization, indexed debt conversion, and share rounding remain part of that check. The zero residual-exposure field is an accounting result of reconstruction, not a guarantee against economic loss.

4.2 Funding recovery and terminal closeout

Let D be actual indexed funding debt, A the canonical opposite claim, and Y revenue already available for netting. The recovery gap is

$$g = [D - Y - A]_+ . \tag{13}$$

For positive A, recovery ignores gaps below 25 basis points of A. Above that threshold the discount is

$$d = \min\left(0.05, \frac{0.8g}{A}\right). \tag{14}$$

It reaches its maximum at $g = A/16$. A spot swap supplying the borrowed asset can match $m = \min(g, \text{input})$. If that matched amount converts to a fair target amount v, the extra target output is

$$\beta = \min\left(E_t, \left\lfloor \frac{vd}{1-d} \right\rfloor\right). \tag{15}$$

The full trade still pays ordinary fees and follows the ordinary curve. The bonus consumes the selected hLP's equity. Critical stress is reported on actual debt before revenue netting, at $D/A \geq 9/8$; it does not gate ordinary swap recovery. Swaps remain available in reduce-only mode and require the caller's input. The atomic flash recovery accepts a positive funding gap, or a frozen vault, before that boundary. It fixes a net opposite-token payment, transfers target collateral at an affordable discount, and requires same-transaction settlement. A full flash close remains subject to the terminal insolvency checks.

Terminal hLP closeout is narrower than arbitrary debt forgiveness. It requires insolvency at both the executable curve mark and symmetric EMA, retires the vault's yLP exposure,

and allocates recovery to principal first. Bounded insurance covers eligible principal shortfall; remaining principal loss is written off and unpaid interest is canceled. Previously funded holder yield claims remain recorded. This is separate from ordinary borrower liquidation and from a routine hLP withdrawal.

4.3 Leveraged liquidity and Yield Basis

The constant-product LP convexity argument explains why two-times leverage is a useful reference point [5]. It does not prove that every concentrated LP portfolio has that leverage, or that funding and trading losses cancel. Yield Basis maintains leverage around externally held Curve LP collateral and crvUSD debt [6]. Dusk instead matches opposite-asset claims and debt inside a pair-local endpoint reconstruction. Neither architectural description establishes a ranking of returns, costs, or resilience.

Dusk's target-asset equity can change through funding interest, fee allocation, recovery incentives, settlement, and market losses. A neutral opposite-asset claim at an admitted endpoint removes one accounting mismatch. It does not make target equity constant across every operation. In particular, equation (15) makes the cost of attracting recovery flow explicit: the trader's additional output is a reduction in hLP equity.

5 Credit, Rates, and Liquidation

5.1 Borrow admission

Public credit uses a shadow constant-product risk curve derived from the full-range tail, conservative depth, and internal price references. In a collateral-to-debt orientation, write

$$\widehat{p} = \min(p_{\text{symEMA}}, p_{\text{dirEMA}}), \qquad L_{\text{hist}} = \min(L_{\text{observed}}, L_{\text{EMA}}). \tag{16}$$

When one historical depth observation is zero, the nonzero observation is used. Write $L_{\text{total}} = L_0 + L_{\text{concentrated}}$ for current prepared center depth. The shadow liquidity scales only the tail:

$$\widehat{L}_0 = L_0 \frac{\min(L_{\text{hist}}, L_{\text{total}})}{L_{\text{total}}}, \tag{17}$$

with $\widehat{L}_0 = L_0$ when historical depth is unavailable. The conceptual risk reserves are $\widehat{x} = \widehat{L}_0/\sqrt{\widehat{p}}$ and $\widehat{y} = \widehat{L}_0\sqrt{\widehat{p}}$. Impact-aware value is

$$W(c) = \frac{c\widehat{y}}{\widehat{x} + c}. \tag{18}$$

The full concentrated execution curve is not simply substituted into borrow admission. The public-borrow book also maintains debt-capped aggregate underwriting contributions; those aggregate contributions do not lock other accounts' idle collateral.

A successful public borrow recomputes and stores the position's liquidation collateral factor, subject to the borrower's minimum acceptable factor. Borrow capacity uses a five-percent

buffer below that factor. On collateral withdrawal the check uses the greater of the stored and newly computed factor, then tests the post-withdrawal buffered capacity. Full repayment clears the factor. Each debt side is secured by that position's opposite-asset collateral.

Public cash borrowing additionally consumes a leaky 24-hour budget derived from conservative depth. The default limit is 20 percent and the configurable ceiling is 30 percent of the relevant depth measure. Only new gross public borrow consumes this budget; repayment does not refill it immediately. hLP funding and isolated leverage do not consume this particular public-borrow bucket, although both face their own checks and enter utilization.

5.2 Adaptive interest

For each asset, utilization includes all indexed debt without double counting:

$$u_i = \frac{D_i^f + D_i^\ell + D_i^h}{D_i^f + D_i^\ell + D_i^h + C_i}. \tag{19}$$

The default target is $u^* = 0.70$, with normalized error

$$e(u) = \begin{cases} (u - u^*)/u^*, & u \le u^*, \\ (u - u^*)/(1 - u^*), & u > u^*. \end{cases} \tag{20}$$

For curve steepness κ, the instantaneous APR is $r = r_T c(e)$, where

$$c(e) = \begin{cases} 1 + (\kappa - 1)e, & e \ge 0, \\ 1 + (1 - 1/\kappa)e, & e < 0. \end{cases} \tag{21}$$

The default $\kappa = 4$ gives multipliers from 1/4 to 4. Dusk uses the adaptive-anchor idea associated with Morpho's AdaptiveCurveIRM [4], with a bounded linear update

$$r_T' = \mathrm{clip}\left(r_T\left[1 + \mathrm{clip}(se\tau, -0.5, 0.5)\right], 0.001, 2\right), \tag{22}$$

where τ is elapsed time in years, capped at one year per update, and default speed is $s = 20$. The anchor starts at four percent APR. Its 0.1–200 percent bounds apply to the anchor, not the final borrower APR after multiplication.

The program accrues the borrow index at the rate computed from the pre-update state, then updates the anchor. Elapsed time is inferred from slots at 400 milliseconds per slot. Debt and index arithmetic round in integer fixed point, so continuous-time formulas or a particular update cadence must not be presented as bit-for-bit execution under arbitrary activity.

5.3 Liquidation and isolated leverage

Public liquidation eligibility uses debt versus the position's stored liquidation factor and a linear mark at the symmetric price EMA. Auction pricing is a different calculation: the floor is the average full-position concentrated unwind value using the symmetric reference price and

pessimistic depth. The auction starts at a five-percent premium to that floor and decays over five minutes. External fillers repay in the debt asset and receive collateral under the auction terms.

After expiry, the backstop can unwind through the live curve, pay a 0.5-percent bounty on consumed collateral, apply capped insurance, and socialize residual loss in the market. Surplus debt-asset proceeds return to the borrower. Insurance has hard concentration ceilings of 20 percent per event and 50 percent per slot-based day window; governance can impose lower limits. These insurance limits are not a guarantee that LP losses cannot exceed a fixed percentage.

Isolated spot-margin leverage has its own collateral and debt shares. In an idealized opening, margin m and multiplier $\mathcal{L}$ imply borrowing $m(\mathcal{L}-1)$ and swapping the resulting notional through the market. Actual entry pays fees, impact, and applicable transfer costs. Initial and maintenance health use position closeout value and unwind constraints, rather than public-borrow aggregate contributions. Increases, decreases, closeout, and liquidation reuse the concentrated swap and cash settlement machinery. A conditional order or liquidation trigger establishes permission to attempt execution, not a guaranteed fill.

5.4 Conditional execution and market controls

The leverage-delegate program supports delegated take-profit and stop-loss closeouts, escrowed leverage entries with limits and expiry, and hLP stop orders. hLP stop-loss uses principal NAV per share; its stop-rate condition uses a fixed twelve-hour EMA of funding APR. An hLP order preserves the original owner's yield destination while shares are escrowed. Execution reserves a five-percent output bounty as a net executor credit, rounded up, then applies the owner's net payout bound; the protocol service fee uses any remaining budget. Entry cancellation and execution can require a minimum actual owner refund credit. Residual published yield can be settled after the order leaves escrow.

Parameter governance exposes seven families: fees, concentration, interest, EMA half-lives, public daily borrow limits, the center controller, and insurance draw caps. Proposal timing and execution conditions are part of the implementation. Market start state, liquidity gates, and reduce-only controls can constrain individual instructions. Their presence describes available control mechanisms; it does not establish a production deployment or a release policy.

6 Numerical Illustrations and Limitations

The companion generator replaces the old synthetic-depth and discrete pre-rebalance experiments with three illustrations of the current mechanism. It produces the tables and vector figures used by the two drafts. Inputs are disclosed in the generated metadata. The curve illustration holds a chosen geometry fixed; the rate illustration updates once per day at fixed utilization; recovery examples use the integer discount and bonus rules. None runs the complete Solana program.

Fixed utilization	Day-30 anchor APR	Day-30 borrower APR
0%	0.7376%	0.1844%
70%	4.0000%	4.0000%
100%	19.8196%	79.2786%

The rate rows use $u^* = 70\%$, $\kappa = 4$, $s = 20$, and an initial four-percent anchor. The concentrated example uses a positive tail and equal core/shoulder layer liquidities to display the five regions. It is an illustrative geometry, not a recommended deployment configuration. Recovery figures distinguish the 25-basis-point activation threshold, maximum discount at a gap of 1/16, and the separate critical test on actual debt.

Several limitations follow from the model. Cash exhaustion can prevent an otherwise valid quote or withdrawal from settling. Internal price filters can lag or be manipulated, and a positive full-range tail does not guarantee useful size at every price. Funding growth can erode hLP equity even when opposite claims are reconstructed correctly. Auction participation, rescue flow, and permissionless terminal calls remain liveness assumptions. Insurance and explicit loss accounting determine how losses are allocated; they do not supply unlimited capital.

The implementation also has finite arithmetic precision, state-dependent fees, and operation ordering that a continuous derivation cannot certify. The most useful next empirical work is exact-program replay across cash stress, boundary crossings, funding recovery, and liquidation, with fee and funding costs included. Historical prototype benchmarks should not be used as measurements of this source snapshot. This revision reports no throughput, return, or comparative performance result.

Reproducibility

Both papers share the same mechanism source so that corrections to equations and parameters remain consistent. `paper/simulations/generate.py` generates the numerical data, summary table, and vector figures. `paper/SOURCES.md` maps claims to implementation paths and describes the verification scope. The expanded appendices provide derivations, worked examples, and a transition-level review map. Code remains the authority for rounding, account validation, instruction ordering, and rejection conditions.

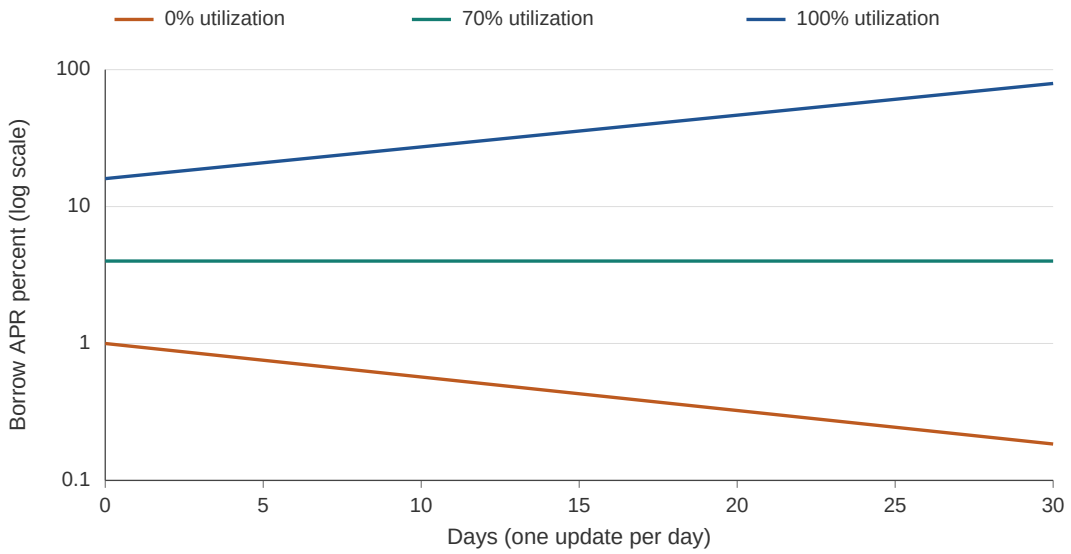


Figure 2: Fixed-utilization rate paths with one anchor update per day and current default parameters. Values show the instantaneous APR after each update. This cadence is part of the experiment, not a claim of cadence-independent onchain accrual.

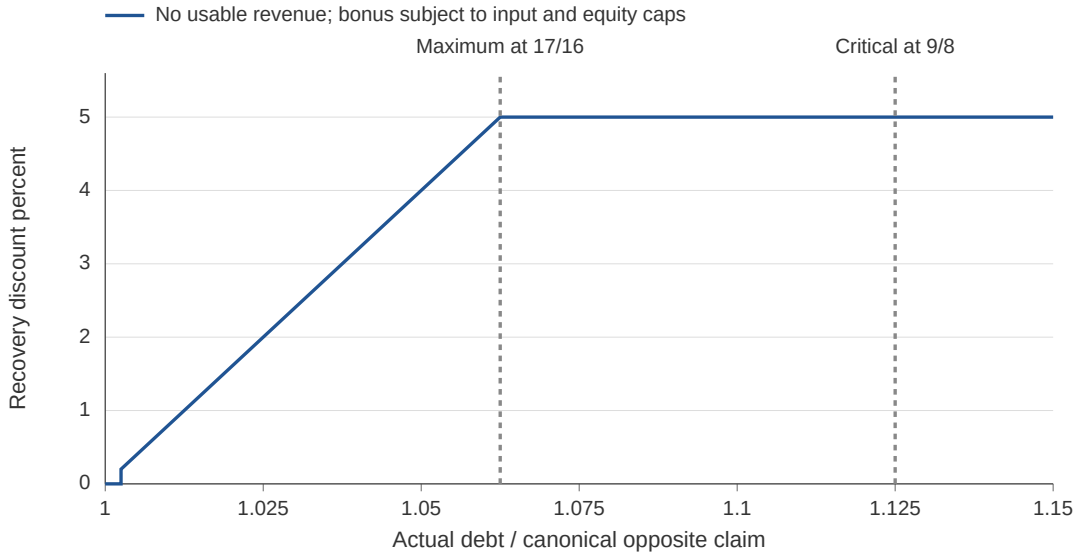


Figure 3: Recovery discount with no usable revenue. Maximum discount occurs at debt/claim 17/16; critical stress begins at 9/8. The plotted fine-scale discontinuity at 25 basis points reflects the activation guard. Bonuses also depend on matching input, conversion ratio, integer rounding, and available equity.

A Reserve Coordinates and Worked Accounting

The following notation distinguishes five objects that were collapsed in the earlier constant-product draft.

Object	Role
V_i^{reserve}	Physical token balance; custody for several separately owned buckets.
C_i	Accounted reserve cash; subject to instruction-specific availability checks.
R_i	Stored live reserve satisfying the cash/debt/hLP identity; basis of proportional yLP mint and burn amounts.
$X_i = R_i - J_i$	Executable reserve coordinate after removing unpaid public and isolated interest.
(U, V)	Ordinary yLP tranche after separating both hLP ownership claims; the trader-facing curve coordinates.

Neither the physical vault balance nor the stored live ratio is sufficient to reconstruct a concentrated swap quote. A quote also requires curve geometry, the ordinary tranche, applicable fees, cash policy, and the prepared hLP transition. This does not make equation (1) optional: every successful instruction must still account for the resulting stored reserve.

A.1 A public loan without hLP

Start with a constant-product market whose two accounted cash reserves are 1,000, with no debt, hLP, or unpaid interest. Let the quote-side borrower take 100. Ignoring share rounding, the quote-side ledger becomes

$$C_q = 900, \quad D_q^f = 100, \quad P_q^f = 100, \quad R_q = X_q = 1{,}000.$$

Base remains unchanged. The principal borrow has changed cash composition without changing the curve point. Utilization is now $100/(900 + 100) = 10\%$ on the quote side, before any other debt or cash changes.

Suppose quote debt later accrues to 110. The ledger becomes $R_q = 1{,}010$, but $J_q = 10$ and $X_q = 1{,}000$. A quote made from raw R_q would count ten unpaid interest tokens as new curve principal. The implementation's subtraction prevents that error. Actual token cash remains 900 until another operation moves tokens.

A full repayment transfers 110 to the settlement path. The principal portion 100 replenishes C_q; the interest portion 10 follows interest distribution. Afterwards $D_q^f = P_q^f = J_q = 0$, $C_q = R_q = X_q = 1{,}000$. There is now realized yield, but it should not be added a second time to executable principal. Any later harvest settles its own published liability.

This example is intentionally independent of hLP. Once hLP is present, the same debt/principal distinction remains, while the integrated transition separately updates vault funding, ownership, and interest. Cash-backed interest and hLP funding interest must not be combined into one undifferentiated reserve increment.

A.2 A swap after the borrow

Return to the state immediately after the 100 quote borrow. A fee-free input of 100 base into the ordinary constant-product curve gives

$$\Delta q = \frac{100 \cdot 1{,}000}{1{,}000 + 100} = 90.90909\ldots.$$

The ordinary and executable reserves become $(1{,}100, 909.09091\ldots)$. Quote cash is $809.09091\ldots$, while quote debt remains 100. Thus $R_q = C_q + D_q^f$ still holds. A larger theoretical curve output would remain subject to the actual cash available for its payout.

If a fee is withheld, its destination matters. A claimable fee stays in custody outside executable reserves; a compounded LP fee enters principal through its allocation transition; retained dynamic surcharge becomes protected recenter funding. A token transfer fee can make actual reserve credit smaller than the user's gross debit. Each of these changes a different term in the accounting, so a single formula of the form "vault balance equals AMM reserve" is insufficient.

A.3 Published yield and share changes

For a growth index stored at scale 2^{64}, a user's incremental whole-token liability has the form

$$\left\lfloor \frac{s(G - G_0) + \rho}{2^{64}} \right\rfloor,$$

where ρ represents the carried fractional contribution under the relevant accounting helper. Global distribution and account-level checkpointing have distinct remainders. The point is to preserve previously earned fractions and allocate newly distributed value to the eligible supply; it is not to promise that each intermediate floor disappears.

When a holder burns principal shares, prior published yield is checkpointed before its balance changes. After terminal hLP closeout, remaining hLP tokens may represent zero principal while already published yield remains claimable. A preview or accounting dashboard should report these components separately. A single aggregate NAV number can otherwise hide a surviving liability or imply an unavailable cash payout.

B The Nested Curve as a Piecewise Analytic Path

For one bounded range $[a, b]$ in square-root-price space, its inventory is

$$x_{[a,b]}(z) = \begin{cases} L(1/a - 1/b), & z \le a, \\ L(1/z - 1/b), & a < z < b, \\ 0, & z \ge b, \end{cases} \tag{23}$$

and

$$y_{[a,b]}(z) = \begin{cases} 0, & z \le a, \\ L(z - a), & a < z < b, \\ L(b - a), & z \ge b. \end{cases} \tag{24}$$

Each inventory is continuous at both boundaries. The nonzero full-range tail adds L_0/z and $L_0 z$. Summing two nested ranges yields equation (7). It gives a continuous path even when active liquidity changes abruptly at a boundary.

Within a region, derivatives are

$$\frac{dU}{dz} = -\frac{L_A}{z^2}, \qquad \frac{dV}{dz} = L_A, \qquad -\frac{dV}{dU} = z^2 = p.$$

This establishes marginal price without assuming that price equals the raw inventory ratio. Inactive ranges retain one asset, and active ranges have offsets from their finite boundaries. Those offsets contribute to inventory value but do not alter the derivative ratio inside the region.

B.1 Boundary crossing

For base input, let the next lower square-root boundary be $a < z$. The amount to reach it is

$$\Delta U_{\text{boundary}} = L_A(1/a - 1/z).$$

If the remaining input is smaller, equation (8) finishes the quote. Otherwise the quote consumes this amount, credits the output $L_A(z-a)$, changes the active set, and continues. Quote input proceeds toward the next upper boundary with input $L_A(b - z)$. The full nested path contains only four boundaries, so at most five segments are traversed.

The Rust cache uses normalized integer quantities and precomputed branch geometry. Its construction and rounding need separate review from these real-valued expressions. In particular, the continuous equation is not an instruction to convert arbitrary raw token amounts directly to floating point in an SDK. Exact quotes should follow the program's normalization, direction, and conservative amount rounding.

B.2 A transparent illustration

The generator uses $L_0 = 100$, core liquidity 450 on $[0.95, 1/0.95]$, and shoulder liquidity 450 on $[0.85, 1/0.85]$. These are explicitly chosen square-root-price bounds. They are not raw governance width values and are not intended as a recommended market configuration. At $z = 1$, $U = V = 190$ and active liquidity is 1,000. Outside both ranges active liquidity is 100.

Consequently, center inventory can be much smaller than active curve liquidity, while the tail still holds inventory beyond the concentrated region. This is the geometric source of concentration. It is distinct from adding synthetic hLP claims to a trader's quote. The example plots inventory ratio V/U beside marginal price z^2 to make their separation observable away from the center.

The implementation's peak-amplification control measures center depth relative to a constant-product pool at corresponding center reserves. It should not be read as an arbitrary multiplier of every trade's output, every price region's depth, or every collateral value. Larger trades can cross into lower-depth shoulders and tails, and lending admission uses the tail-based conservative risk construction.

B.3 Controller economics

A change in the concentration center changes the geometry supported by a given inventory. The controller therefore needs both a target and an admission budget. Dusk retains eligible dynamic surcharge in a protected reserve bucket and evaluates a deferred target against the state in which it will be committed. Deposits, withdrawals, losses, and prior trades can invalidate an old cost estimate.

The ordering matters economically. First commit a previously admissible target, then price a user action from the resulting frozen state, and only then use the successful action to update future observations. This prevents the current action from unconditionally choosing a new center based on its own final price. It does not eliminate manipulation through a sequence of transactions or establish that every admitted movement is profitable under an external benchmark.

C Algebraic hLP Ownership

Let base hLP own the fraction E_b/U of the ordinary portfolio, measured before converting that fraction into integer shares. It therefore owns base claim E_b and quote claim E_bV/U. Matching the quote claim with quote funding debt leaves base equity E_b. Quote hLP has the symmetric construction. Adding ordinary and vault-owned inventories gives equation (10).

Factoring the endpoint reveals a useful identity:

$$X_b = U\left(1 + \frac{E_b}{U} + \frac{E_q}{V}\right), \tag{25}$$

$$X_q = V\left(1 + \frac{E_b}{U} + \frac{E_q}{V}\right). \tag{26}$$

Hence $X_q/X_b = V/U$ in exact arithmetic. This is an inventory-ratio identity. It does not equate that ratio with the concentrated marginal price. The inventory ratio is sufficient to match opposite claims; curve geometry remains necessary to price trades and express marked gross leverage.

C.1 A two-vault example

Take ordinary inventory $(U, V) = (1{,}000, 1{,}000)$, base equity $E_b = 100$, quote equity $E_q = 50$, and ordinary supply $S_o = 1{,}000$. Then

$$s_b = 100, \qquad s_q = 50, \qquad S = 1{,}150,$$

and total executable reserves are $(1{,}150, 1{,}150)$. Base hLP's quote claim and funding target are both 100; quote hLP's base claim and funding target are both 50.

Now apply a fee-free 100-base input to the ordinary constant-product tranche, while holding target equities fixed for this illustration. Ordinary output is $90.90909\ldots$ quote and the

endpoint is

$$U' = 1{,}100, \qquad V' = 909.09091\ldots.$$

The new canonical funding targets become

$$D_q^{h,*} = 100\frac{909.09091\ldots}{1{,}100} = 82.64463\ldots,$$

$$D_b^{h,*} = 50\frac{1{,}100}{909.09091\ldots} = 60.5.$$

Total executable reserves are approximately $(1{,}260.5, 1{,}041.73554)$. The trader did not receive output from a 1,150-by-1,150 constant-product pool: quoting the full synthetic total would have given a different result. Reconstruction changes the gross ownership and funding decomposition around the already executed ordinary path.

In integer shares, equation (11) floors the vault balances against the unchanged ordinary supply. This can change total yLP supply even though ordinary holders neither mint nor burn during the swap. Fee eligibility is frozen at the appropriate earlier state, and actual debt shares are converted through the borrow index. A complete transition therefore carries more information than just the real-valued reserve endpoints shown above.

C.2 What the target does and does not hedge

For a base-target vault at a marked price p, gross collateral value in quote units is $E_b p + E_b V/U$. The second term is canceled by opposite debt in the canonical construction. Gross leverage then follows equation (12). At a constant-product point, $V/U = p$ and the ratio is two. At a concentrated point, V/U can be above or below p, and the two target vaults have different gross leverages.

The familiar constant-product LP value proportional to $\sqrt{p}$ provides motivation for continuously adjusted leveraged liquidity. A static two-times LP position exposed to a discrete jump is not the same process as a continuously adjusted position, and neither expression is the current Dusk execution algorithm. The earlier formula $E_0(\sqrt{r} - 1)^2$ should not be used to infer an active Dusk pre-solve threshold or quote correction. The present program quotes ordinary liquidity and reconstructs ownership algebraically.

Funding interest and fees can change the target equities used in reconstruction. Recovery explicitly decreases one target equity. A loss socialized by another path can also affect the economics of the vault's yLP claim. Opposite-asset neutrality at one endpoint therefore does not imply a constant token balance across time or an unconditional impermanent-loss guarantee.

C.3 Settlement preconditions

The integrated engine verifies the expected curve revision, yLP supply, each vault's yLP shares, indexed debt shares, and principal state. It settles the specified funding-interest cash debit, replaces the old hLP live component with the endpoint's funding component, removes current

public and isolated unpaid interest for the endpoint comparison, and verifies the bounded raw-atom distance. It then commits ownership and debt and asserts the reserve ledger.

The current integrated preparation also requires hLP backing inventories to be empty; direct exits account for backing inventory when present. Positive marked NAV alone does not establish that a particular operation is admissible. Available cash, funding-interest settlement, the prepared pre-state, and the instruction's other guards still apply. If these constraints cannot be satisfied, the transaction can fail atomically rather than publishing a partial hedge.

D Recovery Incentives and the Terminal Boundary

Recovery distinguishes three quantities: actual indexed debt D, canonical opposite claim A, and legally usable revenue Y. The net gap is $g = [D - Y - A]_+$. The critical flag instead compares actual D with $9A/8$ before subtracting revenue. These tests answer different questions: the net gap determines the incentive currently needed, while the actual ratio identifies the debt stress boundary.

For $A = 1{,}000{,}000$ raw units, the following examples have no usable revenue and a one-to-one target conversion. Assume matching input is at least the gap and target equity is large enough to pay the bonus.

Debt	Net gap	Discount (bps)	Bonus (atoms)	Critical
1,002,499	2,499	0	0	No
1,002,500	2,500	20	5	No
1,062,500	62,500	500	3,289	No
1,125,000	125,000	500	6,578	Yes

At a five-percent discount, an amount with fair target value v buys total target value $v/0.95$, so the incremental bonus is $v(0.05/0.95)$. Multiplying by 0.05 alone understates the bonus. The implementation floors the conversion and then the discount arithmetic, caps matched input at the gap, and caps the bonus at available target equity. These caps do not remove the ordinary curve impact or fees on the complete input.

D.1 Revenue can change the incentive without changing the critical flag

For example, let $D = 1{,}130{,}000$, $A = 1{,}000{,}000$, and usable revenue $Y = 120{,}000$. Actual debt exceeds $9A/8$, so the critical flag is true. The net gap is only 10,000, giving an 80-basis-point discount before the other caps. If usable revenue covers the entire excess, the net gap is zero and no positive recovery bonus is produced, even though the actual-debt critical test remains true. The flag is diagnostic; an ordinary swap can earn a bonus whenever the net gap and matching input support one.

When the claim is zero, actual debt can still imply critical stress, but ordinary recovery pricing returns no positive incentive from the ratio formula. This avoids dividing by zero or using

an undefined conversion to fund an output. Terminal closeout and the broader state checks, rather than an extrapolation of equation (14), determine the admissible next action.

D.2 Terminal closeout is an interest-loss path

The terminal engine checks that marked collateral is below indexed funding debt and that the shortfall is no larger than the unpaid interest. This distinguishes passive funding insolvency from a principal hole. The plan is bound to the state it valued, records the permitted insurance request, retires the vault's owned yLP, and realizes the recoverable interest. Insurance is senior to the remaining socialized interest loss and is constrained by available capital, protocol concentration limits, and caller bounds.

The caller also supplies a maximum socialized-loss bound and minimum bounty output. The bounty is computed from recovered funding interest; it is not an extra source of recovery capital. Published user yield remains a separate liability. A surviving hLP token balance can be burned for zero principal after a terminal event, while its checkpointed yield can still be harvested.

This mechanism provides an explicit action once funding growth has exhausted equity, but depends on someone submitting an admissible transaction. A bonus or bounty does not prove that an economically motivated caller will exist for every token, especially when transfer costs, fees, congestion, or small amounts dominate. These are liveness questions for stress testing and operational observation.

E Three Valuations in Public Credit

Borrow admission, liquidation eligibility, and liquidation pricing deliberately use different functions. Keeping their roles separate prevents an appealing but incorrect statement that "all risk uses the same AMM output formula."

Decision	Valuation role
New public credit	Conservative tail-based shadow CPMM, lower directional/symmetric collateral price, pessimistic depth, position and aggregate underwriting checks.
Liquidation eligibility	Linear collateral mark at symmetric EMA compared with indexed debt and the position's stored liquidation collateral factor.
Auction floor	Average full-position concentrated unwind at the symmetric reference and pessimistic depth.
Expired backstop	Actual live-curve unwind with real cash, fees, transfers, insurance, and residual loss settlement.

E.1 Admission and collateral factors

Equation (18) is concave in collateral size. Its derivative is

$$W'(c) = \frac{\widehat{x}\widehat{y}}{(\widehat{x}+c)^2}, \qquad W''(c) = -\frac{2\widehat{x}\widehat{y}}{(\widehat{x}+c)^3} < 0.$$

It recognizes less than the linear mark $c\widehat{y}/\widehat{x}$ for every positive collateral amount. That property alone does not prove that dividing collateral among accounts cannot expand total borrowing: a concave function applied separately can produce a larger sum. The market-level underwriting contributions and global checks must also be considered. The paper therefore does not claim account-splitting resistance solely from the one-position formula.

A borrow stores a newly computed liquidation collateral factor f. The user can specify a minimum acceptable value; borrow capacity is buffered below the accepted factor. In fraction notation, the buffered factor is $0.95f$. On collateral removal, the implementation takes the greater of the stored factor and the newly calculated factor before checking remaining collateral. These are action-specific rules, not a continuously reset liquidation factor after every market movement.

Each public debt side uses the borrower's opposite-asset collateral. The aggregate health book recognizes a debt-bearing contribution capped by underwriting logic, rather than claiming that all idle collateral across accounts is locked for other borrowers. Removing a contribution from a global summary does not move another account's tokens. Exact debt-share rounding and contribution updates are important coupled-state checks in a source review.

E.2 Observation timing and the public borrow budget

An internal EMA can be represented conceptually as $m' = \alpha m + (1-\alpha)x$, with decay determined by elapsed slots and a half-life. Dusk has separate observations for price, directional price, curve depth, and concentration center. Their role is to avoid treating every instantaneous favorable movement as immediately recognized borrowing power. They cannot establish that the market has truthful external information.

The public borrow limiter is a continuously draining usage bucket over a 24-hour window, not a UTC-midnight reset. Gross new public principal consumes its available budget. Debt repayment restores cash but does not immediately undo gross usage. The implementation carries fractional leakage so repeated updates do not lose all sub-unit drain. This property is specific to the limiter and should not be generalized to every fixed-point mechanism in the protocol.

E.3 Auction progression

Once the stored-factor trigger is met, the auction snapshots its reference floor and starts five percent higher. The auction's time progression is measured in timestamps over five minutes. A fill repays debt tokens in exchange for collateral under the current auction terms. The auction

price is not an external fair-value assertion; it is anchored to the market's own conservative recovery estimate.

After expiry, the backstop uses a live unwind and its realized output. The collateral bounty is taken in the documented settlement path, insurance is drawn subject to its concentration budgets, and remaining loss is socialized automatically by that path. A generic assertion that every liquidation requires an arbitrary user-selected loss cap would be inaccurate. Terminal hLP closeout has explicit caller caps; the public expired-auction path has its own settlement semantics.

The hard insurance ceilings are upper bounds on how much insurance one event or day window can consume, not upper bounds on the size of a debt deficit. A market can lose more than insurance covers. When describing resilience, insurance availability, residual LP loss, and the ability to execute a liquidation should be reported separately.

F Rate Controller and Update Cadence

The implemented defaults and accepted governance range are:

Parameter	Default	Range or bound
Target utilization	70%	60%–75%
Curve steepness κ	4	2–8
Anchor adjustment speed	20/year	1–50/year
Initial anchor APR	4%	Initialization value
Anchor APR	Adaptive	0.1%–200%
Anchor multiplicative step	State-dependent	Factor 0.5–1.5 per update
Accrued elapsed time	Slots $\times$400 ms	At most one year per update

With the default steepness, a four-percent anchor implies one-percent APR at zero utilization and sixteen-percent APR at full utilization. At the maximum 200-percent anchor, full utilization gives 800-percent instantaneous APR under the default steepness. The anchor bound cannot be advertised as the user's maximum borrowing rate. Governance changes to steepness change this multiplier range.

The implementation first computes utilization and the instantaneous rate using debt, cash, and the current anchor. The index increment has the form

$$I' = I + \left\lfloor \frac{I \lfloor r_{\mathrm{NAD}} \Delta t / T \rfloor}{\mathrm{NAD}} \right\rfloor,$$

where T is milliseconds per year and Δt is capped elapsed time. It then adapts the anchor with equation (22). Fixed/public and isolated indexed debt are converted separately; summing shares before a single conversion would change some raw-atom boundaries. hLP funding debt uses the same asset-side index but has different live-reserve treatment.

The daily illustration uses $\tau = 1/365$ for thirty updates and holds utilization fixed by assumption. It records the APR after each anchor update, not a borrower's realized annual return. Fixed utilization would itself require compensating supply or debt changes as interest accrues; the example isolates the controller rather than simulating those agents.

For comparison, collapsing thirty days into a single update at full utilization gives $r'_T = 0.04(1 + 0.5) = 0.06$ because the step is clamped. Thirty daily updates produce a different anchor. Even below the clamp, repeated linear factors and a single larger linear factor are not identical. This makes update cadence a disclosed experiment parameter. It also motivates exact-program testing of very frequent and very sparse activity rather than inferring frequency independence from the controller's intended exponential intuition.

With no debt shares, the program moves the anchor toward its lower bound and advances the accrual clock, without generating debt interest. When hLP debt exists, funding APR observations include its current asset-side rate and update the vault's twelve-hour funding-rate EMA. That signal is also used by hLP conditional stop-rate orders, tying the order threshold to a smoothed funding measure rather than the instantaneous anchor alone.

G Settlement, Delegation, and Economic Boundaries

G.1 Fee basis and token transfers

Let a be the amount credited to reserve custody after an input token transfer. For input-denominated fees, the total Dusk fee budget is at most $\lfloor a/2 \rfloor$; executable input is reduced by the priced fee. For an output-denominated fee, the prepared curve produces gross output b and the fee budget is at most $\lfloor b/2 \rfloor$. User output is reduced by the withheld fee and then by any token-program output transfer effect.

These definitions matter when quoting a minimum received amount. A bound on protocol fee share does not bound AMM price impact. It also does not erase a Token-2022 transfer fee. A leverage entry limit uses the all-in relation between paid notional and collateral actually obtained, with direction-appropriate rounding. A client that treats gross output, fee-net output, and final token credit as the same amount can accept a worse trade than the user intended.

Fees also need correct ownership timing. The trade's eligible ordinary supply and vault-owned yLP balances are captured before later reconstruction changes those balances. Claimable yield is published through its liability indexes. Compounded yield increases principal and changes the target equities through the integrated fee allocation. Protected recenter funds remain a separate custody claim. These are three different destinations for collected value, each with a different effect on future quotes and redemptions.

G.2 Leverage and delegated orders

Public borrowing leaves the principal curve coordinate unchanged at its cash/debt step. Isolated leverage composes that step with a real swap. This makes entry immediately sensitive

to depth, fees, and cash availability. The resulting collateral and debt belong to a dedicated position, and its closeout value is recomputed from executable market machinery. An increase or margin withdrawal can fail if initial health or unwind-impact checks are not met.

The delegate program adds conditional authority around Dusk instructions. A take-profit or stop-loss condition, a leverage-entry price limit, or an hLP funding-rate threshold narrows when an executor can act. The underlying market instruction still validates its accounts and settlement. Conditional permission is not reserved liquidity, guaranteed execution, or a promise that a threshold crossing will be observed in time.

hLP orders escrow the hLP shares while preserving the original owner's yield recipient. On execution, the output bounty is charged before evaluating the configured owner payout floor; final transfer fees remain relevant. After cancellation or execution, the dedicated yield-settlement path can harvest remaining liabilities and close the appropriate order state. The escrow's token ownership should not be interpreted as transferring economic entitlement to already assigned user yield.

G.3 Owner, recipient, and harvest authority

The optional harvest authority is stored separately from the yield recipient. The owner can grant or revoke it without changing where funds go, and can change the recipient without implicitly changing the independent authority. Harvest authorization accepts the owner, recipient, or configured authority. Account validation restricts payout to the recipient's canonical associated token account.

This division is useful for program-owned positions and automated collection. It is also a source of documentation error if described simply as "permissionless harvest" or "only the owner may harvest." The current rule is an explicit set of authorized signers with a constrained destination. A program-derived owner still needs its program to sign the invocation for owner-only configuration changes.

H Implementation Evidence and Reproducibility

The review snapshot is commit `263c1635ad87`. The complete identifier is recorded in `paper/SOURCES.md` and generated data metadata. The table below identifies semantic entry points; it is not a substitute for the surrounding account checks and caller ordering.

Claim	Implementation anchor (under `programs/dusk/src/`)
Stored reserve identity	`transitions/ledger.rs`: `assert_virtual_reserve_invariant`.
Unpaid-interest exclusion	`transitions/amm/mod.rs`: `unrealized_interest`, `curve_reserve`.
Nested curve	`transitions/amm/curve.rs`: geometry preparation and analytic segment quotes.
hLP ownership	`transitions/liquidity/hlp/integrated.rs`: `reconstruct_hlp_endpoint`, `reconstruct_hlp_ownership`.
hLP settlement	`transitions/liquidity/hlp/engine.rs`: identity-bound transition, cash floors, terminal waterfall.
Recovery discount	`transitions/liquidity/hlp/recovery.rs`: `quote_hlp_recovery`.
Rates and debt accrual	`math/risk.rs`; `transitions/lending/mod.rs`: `accrue_side`.
Public auctions	`instructions/lending/liquidation/`: start, fill, backstop, settlement.
Isolated leverage	`transitions/leverage.rs`; corresponding instruction handlers.
Custody and harvest	`instructions/accounts.rs`; `instructions/liquidity/harvest.rs` and `state/yield_account.rs`.

Some functions and fields retain historical names. A helper called "live," "rebalance," or "tracking" is not evidence that the earlier paper's algorithm remains in use. The review followed the active callers and distinguished test-only helpers from executable paths. The old tracking-loss solver and generic test-only fee routines should not be used to describe current integrated swaps.

H.1 Rebuilding the illustrations

Run the generator from the repository root with a Python environment containing ReportLab. Its arithmetic uses the Python standard library; ReportLab is used only to create vector charts. The script writes CSV data, machine-readable assumptions, two LaTeX tables, and three PDF figures. It checks range-boundary continuity, rate anchor bounds, recovery thresholds, and selected accounting examples before writing its validation receipt.

Compile either top-level LaTeX source from the `paper/` directory using Tectonic. Both files include the same `main.tex` and `preamble.tex`; the expanded version enables additional discussion and includes these appendices. This organization prevents a parameter correction in one paper from leaving the other with an incompatible equation. The bibliography contains only sources actually used in this revision.

The numerical checks establish properties of the disclosed illustrations. They are not differential tests between Python and every Rust instruction. The source review and the repository's Rust tests provide additional implementation evidence, recorded separately in the verification receipt. This documentation update makes no changes to program logic or SDK interfaces.

H.2 What a stronger evaluation would measure

A useful replay should run the exact current program through multiple boundary crossings, both fee denominations, asymmetric token decimals, funding accrual, hLP recovery, and public auction expiry. Report cash by side, debt by domain, ordinary reserve depth, total reserve claims, hLP target equity, interest paid, insurance draw, and socialized loss. Net economic returns require trading fees, funding cost, execution costs, and the chosen external benchmark, not just a reserve-identity check.

Adversarial sequences should include favorable price manipulation before borrowing, temporary depth provision, collateral withdrawal after a stored-factor update, many small accrual intervals, scarce cash during hLP interest settlement, and concurrent fee compounding with a large swap. A review should distinguish a rejected transaction from a successful but loss-making one. Rejection can preserve accounting while reducing market availability; a successful transaction can preserve every identity while transferring economic value between participants.

For auctions and recovery, participation must be modeled explicitly. An attractive theoretical price does not ensure a filler has the required asset or can profit after execution costs. Measure time to an admissible fill, remaining collateral, realized output, available insurance, and the residual loss. For hLP terminal closeout, separately verify the interest-only insolvency precondition and preservation of published yield.

No claim of current comparative performance is imported from the earlier constant-product or prototype concentrated-curve studies. Those experiments evaluated different mechanisms. A new comparison would need a common workload, precisely matched capital and risk assumptions, current executable artifacts, and a disclosed source revision for every system. Until then, the external references in this paper provide mathematical and architectural context rather than a performance ranking.

References

[1] Omnipair. Generalized automated market maker (GAMM). https://docs.omnipair.fi/technical-breakdown/generalized-automated-market-maker, 2026. Accessed September 10, 2026. Cited for GAMM lineage, not as authority for this Dusk source snapshot.

[2] Michael Egorov. StableSwap: Efficient mechanism for stablecoin liquidity. https://classic.curve.finance/files/stableswap-paper.pdf, 2019.

[3] Michael Egorov. Automatic market-making with dynamic peg. https://classic.curve.finance/files/crypto-pools-paper.pdf, 2021.

[4] Morpho. AdaptiveCurveIrm.sol. https://github.com/morpho-org/morpho-blue-irm/blob/main/src/adaptive-curve-irm/AdaptiveCurveIrm.sol, 2026. Source accessed September 10, 2026; cited for the adaptive rate-at-target mechanism.

[5] Michael Egorov. Eliminating impermanent loss by leveraged liquidity, June 2025. June 12 manuscript supplied as leveraged-liquidity-paper.pdf; theoretical context.

[6] YieldBasis. YieldBasis technical overview. https://docs.yieldbasis.com/dev/overview, 2026. Accessed September 10, 2026; architectural context, not comparative performance evidence.